Random-Intercept Models

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Introduction

A random-intercept model allows each cluster (subject, hospital, school) its own baseline intercept while constraining all clusters to share the same regression slope. It is the simplest mixed-effects model and the natural starting point for repeated-measures, longitudinal, or clustered data. The single new parameter beyond ordinary regression — the variance of the random intercept — captures between-cluster heterogeneity in baseline level and is the primary handle for partial pooling and intraclass correlation.

Prerequisites

A working understanding of mixed-effects models, the difference between fixed and random effects, and basic familiarity with `lme4` formula syntax.

Theory

The random-intercept model is

$$Y_{ij} = \beta_0 + u_i + \beta_1 X_{ij} + \varepsilon_{ij}, \qquad u_i \sim \mathcal{N}(0, \sigma_u^2), \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2).$$

The cluster-specific intercept $\beta_0 + u_i$ varies around the population intercept β_0 with variance σ_u^2. The intraclass correlation coefficient is

$$\rho = \frac{\sigma_u^2}{\sigma_u^2 + \sigma^2},$$

interpretable as the proportion of total variance attributable to between-cluster differences. A high ICC (> 0.30) indicates substantial cluster-level structure; a low ICC (< 0.05) suggests the random effect is unnecessary and ordinary regression suffices.

Assumptions

Random intercepts are independent and Normally distributed across clusters; residuals are independent and Normal conditional on random effects; all clusters share a common slope (relax via random slopes when this fails).

R Implementation

```r
library(lme4); library(lmerTest)

data(sleepstudy, package = "lme4")
fit <- lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy)
summary(fit)

# ICC
```

```r
vc <- as.data.frame(VarCorr(fit))
icc <- vc$vcov[1] / sum(vc$vcov)
icc
```

Output & Results

`summary(fit)` reports the fixed effects (common slope and population intercept) with `lmerTest`-supplied p-values, plus the random-intercept SD and residual SD. The ICC is computed from the variance components and is the primary single-number summary of cluster-level structure.

Interpretation

A reporting sentence: "A random-intercept model estimated the population-level slope at 10.47 ms per day of sleep deprivation (95 % CI 8.92 to 12.02, $p < 0.001$); between-subject variation in baseline reaction time was substantial ($\hat{\sigma}_u = 37$ ms), with intraclass correlation 0.39 indicating that 39 % of total variance is attributable to between-subject differences." Always report the ICC alongside fixed effects.

Practical Tips

- Always report the ICC; it is the headline measure of cluster-level structure and tells readers whether the random-intercept model was warranted.
- For small numbers of clusters (< 5), random-effect variance estimates are unstable; consider a simpler model with fixed effects per cluster, or move to Bayesian estimation with weakly informative priors.
- Compare against a no-random-effects model via likelihood-ratio test (with the χ^2_0 / χ^2_1 mixture reference for the boundary null).
- Use `confint(fit, method = "profile")` for profile-likelihood confidence intervals on random-effect variances; Wald intervals are unreliable on the boundary.
- If slopes clearly vary across clusters (visible in spaghetti plots), add random slopes via `(1 + X | cluster)`; reporting only random intercepts when slopes vary biases inference.
- For non-Gaussian outcomes, the same syntax extends via `glmer()` with the appropriate family.

R Packages Used

`lme4` for `lmer()`, `glmer()`, and `VarCorr()`; `lmerTest` for Satterthwaite or Kenward-Roger p-values; `performance::icc()` for tidy ICC computation; `pbkrtest::PBmodcomp()` for parametric-bootstrap LRTs; `glmmTMB` for an alternative implementation that handles complex error structures.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)

- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures $\rightarrow$ mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI